

# Quasi-3D Phononic Crystal GRIN Lens for Anisotropic Underwater Acoustic Focusing

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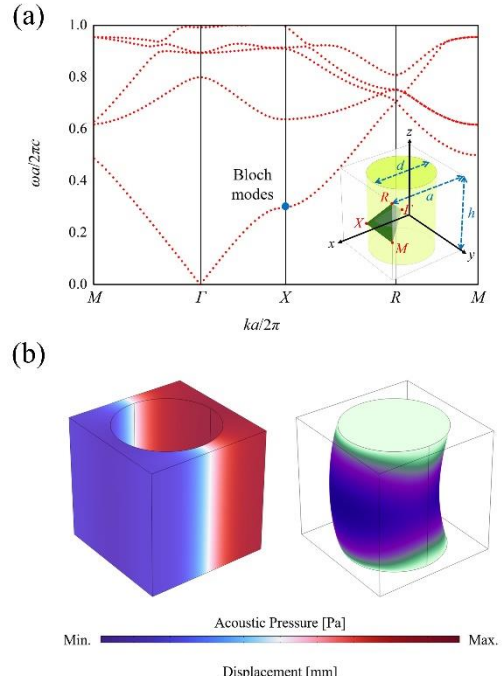
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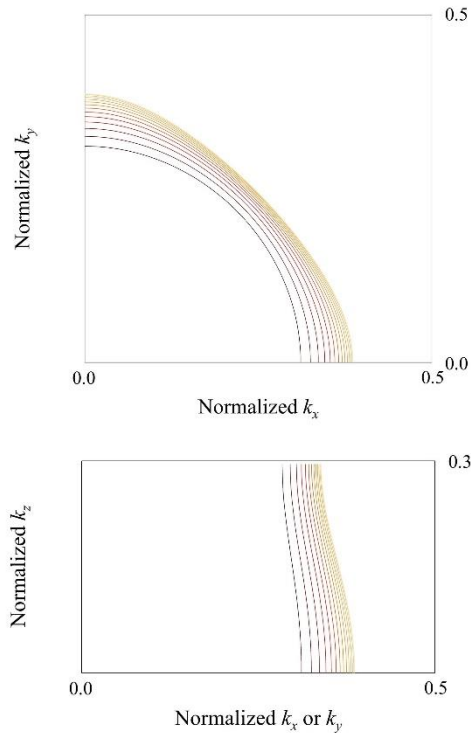
**Abstract:** We propose a quasi-3D phononic crystal GRIN lens for underwater acoustic focusing. Inspired by the Luneburg lens, it features 15 layers of structured meta-atoms, enabling anisotropic wavefront control. Equifrequency contour analysis and experiments confirm elliptical focal formation, distinguishing it from conventional isotropic designs.

Phononic crystals (PCs) and gradient index (GRIN) lenses are widely employed in wave manipulation due to their ability to tailor wavefront propagation through spatially varying refractive indices. GRIN lenses have facilitated precise sound field control and high-resolution focusing. Recent advancements have explored the integration of PC structures within GRIN architectures to achieve enhanced wave steering capabilities. Notably, prior studies have demonstrated attempts to achieve 3D focusing effects using structurally 2D or axisymmetric GRIN designs, leveraging engineered phase gradients or anisotropic effective medium parameters<sup>1</sup>. Fully 3D symmetric Luneburg and ellipsoidal lenses have been investigated for underwater acoustic focusing, offering promising results in omnidirectional beam convergence and wavefront shaping<sup>2</sup>.

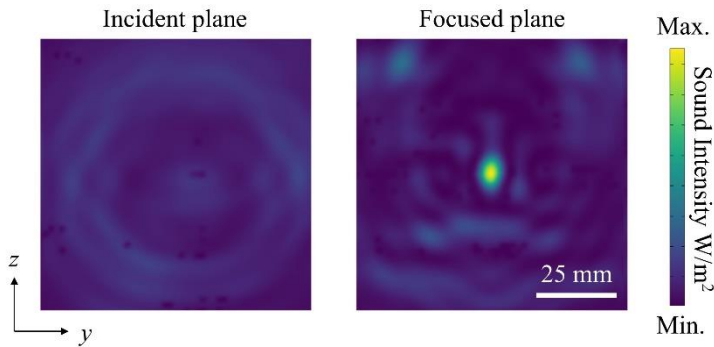
We propose a quasi-3D PC GRIN lens inspired by the Luneburg lens and constructed with 15 layers of structured meta-atoms. The design consists of cylindrical SUS304 meta-atoms embedded in water, forming a GRIN distribution that directs acoustic waves. **Figure 1(a)** shows the band structure, illustrating its dispersion properties. **Figure 1(b)** presents Bloch modes at the X point, revealing localized bending modes that influence wave propagation.



**Figure 1** (a) Band structure of proposed GRIN lens composed of SUS304 (cylindrical meta-atom) embedded in water (cubic unit cell). (b) Bloch modes of the acoustic pressure and displacement at the X point of the band structure, demonstrating bending modes within the meta-atoms.



**Figure 2** EFCs of all meta-atoms at a frequency of 150 kHz, forming the quasi-3D GRIN metalens.



**Figure 3** Experimental results of the quasi-3D GRIN lens focusing performance. Spatial distributions of sound intensity in the incident planes and the focused planes, illustrating the elliptical focal shapes achieved.

The anisotropic nature of the proposed PC is a crucial factor in its quasi-3D behavior. Unlike symmetric lenses, where wave propagation remains isotropic across all axes, our design exhibits directional dependence in the effective wavefront shaping. This effect is confirmed by the equifrequency contours (EFCs) at 150 kHz, shown in **Figure 2**, where the contours deviate from circular symmetry, demonstrating the anisotropic wavefront transformation within the graded structure. The finite height of the meta-atom layers introduces an additional constraint, leading to discontinuities in the energy bands along the vertical axis, a feature not observed in infinite phononic lattices. These discontinuities contribute to the elliptical focal shapes observed in the experimental results.

Experimental validation supports the numerical predictions, showing precise focal tuning. **Figure 3** presents measured sound intensity distributions at the incident and focused planes, confirming the elliptical focal regions. This deviation from a perfect circular focus is a direct consequence of the inherent anisotropy and the finite nature of the system. The results highlight the ability of the quasi-3D phononic crystal GRIN lens to achieve directional focusing while minimizing scattering effects.

By bridging the gap between conventional 2D GRIN designs and fully 3D acoustic focusing, the proposed quasi-3D phononic crystal GRIN lens offers a compact and efficient solution for underwater applications. Future work will focus on broadening operational bandwidths, refining meta-atom geometries, and integrating adaptive tuning mechanisms for applications in sonar imaging, biomedical ultrasound, and underwater sensing.

## References

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- <sup>2</sup> J.-W. Kim, G. Hwang, S.-J. Lee, S.-H. Kim, and S. Wang, *Mech. Syst. Signal Proc.* **179**, 109374 (2022).